

24UITP01	CLOUD COMPUTING AND VIRTUALIZATION			L/T/P/C 2/0/2/3
Nature of Course: Laboratory Integrated Theory				PEC
Prerequisite(s):Basics of Computer Networks				
Course Objectives: Enable the students to				
➤	Analyze the foundational concepts of virtualization and evaluate its role in modern IT infrastructure.			
➤	Analyze and compare various virtualization technologies, architectures, and tools used in computing environments.			
➤	Evaluate containerization technologies and orchestration frameworks, including Docker, Podman, and Kubernetes, for deploying scalable applications.			
➤	Apply open-source virtualization and container platforms to design and implement virtualized and containerized environments.			
➤	Evaluate real-world applications and emerging trends in DevOps, cloud computing, and microservices architectures.			
Course Contents:				
UNIT I	VIRTUALIZATION			CO16
Evolution of Virtualization - Benefits and Limitations of Virtualization - Types of Virtualization - Hypervisors - Virtualization in Modern IT - Types of Virtualization Used in Cloud				
UNIT II	VIRTUAL MACHINE SETUP AND MANAGEMENT			CO26
Introduction to Open-Source Tools: VirtualBox, KVM - Installing and Configuring Virtual Machines - Resource Allocation: CPU, RAM, Disk - Creating and Managing Snapshots - Importing/Exporting VMs, Cloning, and Templates - Practical Challenges in VM Management				
UNIT III	APPLICATIONS AND SECURITY IN VIRTUALIZATION			CO36
Using VMs for Development and Testing - Network Virtualization (NAT, Bridged, Host-Only) - VM Backup and Recovery - VM Isolation and Sandboxing - Security Considerations in Virtualized Environments - Performance Tuning for VMs - Virtualization vs Emulation vs Containerization-Virtualization Security in Cloud				
UNIT IV	CONTAINERIZATION			CO46
Introduction to Containers - Benefits - Docker Architecture: Daemon, CLI, Images, Containers - Installing Docker on Linux/Windows - Basic Docker Commands - Creating Dockerfiles and Building Images - Introduction to Podman as an Alternative to Docker				
UNIT V	CONTAINER ECOSYSTEM AND ORCHESTRATION			CO56
Docker Compose for Multi-Container Applications - Volume Management and Data Persistence - Networking in Containers - Basics of Kubernetes: Pods, Services, Deployments - Setting up Minikube and Writing YAML Files - Container Lifecycle and Security Best Practices - Emerging Trends: Microservices, Serverless Containers, DevOps Integration				
Theory Hours:				30

PRACTICAL EXERCISES:		
1.	Set up VirtualBox and create an Ubuntu virtual machine	CO1
2.	Practice snapshot creation, VM cloning, and export/import	CO1
3.	Configure NAT, Bridged, and Host-only networking for VMs	CO2
4.	Install Docker and run your first container	CO2
5.	Create Dockerfile, build an image, and run a custom container	CO3
6.	Use Docker Compose to run a web and database stack	CO4

7.	Install Minikube and deploy a simple container app with YAML	CO5
8.	Open-Ended Experiment	CO1-CO5
Practical Hours:		30
Total Hours:		60

COURSE OUTCOMES	
At the end of the course student will be able to	
CO1	Analyze a virtual box by applying foundational concepts to distinguish between various types of virtualization and their practical benefits.
CO2	Apply procedures to install, configure, and manage virtual machines using open-source tools like VirtualBox or KVM.
CO3	Analyze the use of virtualization in real-world scenarios such as system testing, deployment, and resource isolation.
CO4	Apply container technologies like Docker or Podman to build, run, and manage lightweight, portable environments.
CO5	Analyze modern orchestration platforms and trends like Kubernetes, microservices, and serverless computing for scalable deployment.

Text Books:	
1	Matthew Portnoy "Virtualization Essentials" 2nd Edition Wiley 2021
2	Nigel Poulton "Docker Deep Dive" 2022 Edition
	Kelsey Hightower "Kubernetes Up and Running" et al., O'Reilly, 2022
Reference Books:	
1	Humble Devassy "Mastering KVM Virtualization" Packt, 2021
2	Gabriel N. Schenker "Learn Docker – Fundamentals of Docker 19.x" Packt, 2020
Online Resources:	
1	https://docs.docker.com
2	https://kubernetes.io/docs
3	https://podman.io
4	https://www.virtualbox.org/wiki/Documentation

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	-	3	-	3	2	2	-	1	3	2	2
CO2	3	3	2	-	3	-	3	2	2	-	1	3	2	2
CO3	3	3	2	-	3	-	3	2	2	-	1	3	2	2
CO4	3	3	2	-	3	-	3	2	2	-	1	3	2	2
CO5	3	3	2	-	3	-	3	2	2	-	1	3	2	2

Correlation levels 1, 2 and 3 are as defined below: 1 - Slight 2 Moderate 3 - Substantial (High)